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Beyond Rule-Based Agents: Active Markov Games for Realistic Multi-Agent Interaction in Autonomous Driving

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Abstract

Current research in autonomous driving heavily relies on large-scale driving datasets for model fitting or trial-and-error learning strategies in simulation environments. However, these approaches suffer from limited behavioral diversity and fail to cover complex edge-case interactions. To address these limitations, we model the driving environment as an Active Markov Game (AMG) and introduce a multi-agent co-evolutionary training framework for more realistic interactive learning. The AMG formulation extends traditional Markov games by explicitly making state transitions and rewards dependent on the evolving strategies of the agents, thus capturing the interactive dynamics and strategic coupling between the ego vehicle and surrounding agents. Building on this, our multi-agent co-evolutionary training mechanism jointly optimizes the ego vehicle's policy and a diverse pool of opponent strategies, allowing all agents to adapt to each other's behaviors during training. This game-theoretic approach produces a robust ego agent capable of handling diverse, non-stationary driving strategies, overcoming the "non-responsive opponent" limitation found in prior methods. In CARLA simulations of unsignalized intersections and long-tail scenarios, our method performs exceptionally well, achieving success rates (98%), and significantly outperforming state-of-the-art baselines such as PPO, DDPG, and IPPO in terms of generalization, safety margins, and control smoothness. These results demonstrate that our approach substantially enhances the robustness, safety, and strategic adaptability of autonomous driving in complex multi-agent environments.

1. Introduction

The decision-making algorithm in autonomous driving systems serves as a crucial component that bridges perception and control, directly influencing the vehicle's safety, efficiency, and adaptability in dynamic traffic environments [26]. However, in complex urban scenarios—particularly unprotected situations such as intersections or roundabouts without traffic lights or signs—navigating through these areas remains one of the major challenges for autonomous driving systems. In such environments, multiple types of traffic participants (e.g., pedestrians, cyclists, and motor vehicles) coexist, exhibiting highly diverse driving behaviors and unpredictable motion patterns, which lead to complex traffic dynamics and significantly increase the difficulty of trajectory planning. In high-density traffic flows, the extremely short inter-vehicle distances amplify interactive effects, imposing stringent requirements on the real-time performance and robustness of the system [33]. Current research on decision-making in complex driving scenarios such as intersections primarily relies on large-scale driving datasets [13, 20, 31, 37] where models are trained to approximate the conditional

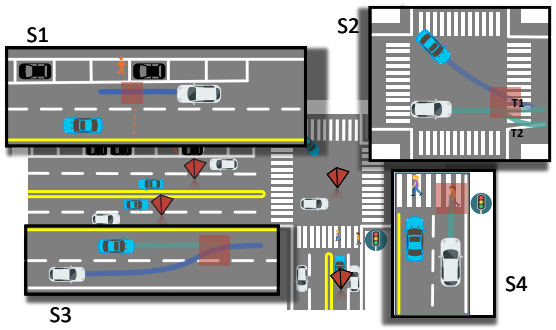


Figure 1. **Illustration of challenging scenarios. S1** and **S3** represent examples of long-tail scenarios that are difficult to cover in the dataset (a sudden pedestrian emergence and a malicious overtaking event); in **S2**, the rule-based agent follows trajectory T1, whereas the interactively trained agent follows trajectory T2; **S4** illustrates a paranoid scenario that may occur in areas lacking traffic infrastructure or where driver discipline is poor.

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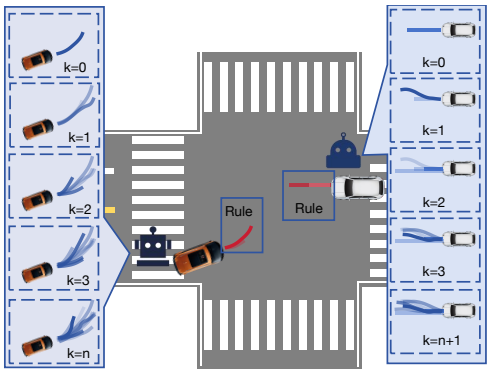


Figure 2. **Motivation.** As the decision-making networks continue to evolve, non-ego agents have developed various interactive strategies targeting the ego vehicle's decisions, thereby realistically reproducing the complex behavioral coupling relationships among drivers in real traffic scenarios.

distributions of existing driving decisions. However, most of these datasets are collected from regions with well-developed traffic infrastructure (e.g., complete traffic light systems and clearly marked road signs) and regulated driving behaviors, lacking sufficient coverage of extreme scenarios, such as those with poor infrastructure, rule violations, or high-risk driving behaviors (e.g., aggressive lane changing or illegal overtaking) [4, 10, 23]. Consequently, in out-of-distribution scenarios, models often suffer from sample sparsity and insufficient causal information, making it difficult to accurately model potential interactions and adjust decisions in a timely manner, thereby compromising overall system safety. To address these limitations, researchers begin leveraging simulation platforms such as CARLA and SUMO [3, 9, 24, 28], incorporating deep reinforcement learning (DRL) to enable models to continuously explore, perform trial-and-error learning, and optimize their strategies in diverse and even extreme dynamic environments [19, 42]. This approach helps mitigate the distributional limitations of real-world datasets and enhances the model's capability to handle complex traffic conditions. However, in most of these studies, other traffic participants are typically simplified as rule-based agents, whose behavioral patterns are predefined and unable to dynamically respond to the ego vehicle's policy. [1, 5, 38] Such modeling weakens the strategic coupling among agents and fails to realistically reproduce the interactive behaviors observed among human drivers in real traffic. Moreover, existing DRL-based methods mainly focus on single-agent optimal policy learning, lacking explicit modeling of multi-agent decision-making interactions, and therefore exhibit limited performance in high-density and strongly coupled traffic scenarios Fig. 1 .

To effectively model the interactive behaviors and potential strategy evolution of agents in dynamic environments, this paper introduces an active Markov game framework [16]. The framework designs the joint policy update function as a function of each agent's action, which influences both state transitions and rewards, thereby enabling the model to effectively capture the interactive dynamics and the temporal evolution of strategies Fig. 2. Furthermore, to endow agents with interactive learning capabilities, we propose a game-theoretic interactive training framework. Unlike conventional deep reinforcement learning methods that optimize a single-agent policy in static environments, our approach adopts a multi-agent co-evolution mechanism. Under this mechanism, each agent leverages the joint policy update function to capture the strategy evolution of other agents and update or refine its own policy accordingly, while maintaining a dedicated policy inference network to approximate and predict the opponents' decision behaviors.

This multi-agent co-evolutionary process can more realistically reproduce the complex behavioral couplings among drivers in traffic scenarios. Consequently, the ego vehicle can accumulate rich interaction experience with diverse and dynamically changing driving strategies, thereby learning robust and generalizable decision-making policies that remain effective under uncertain and variable interactive conditions, and gradually approach a Nash equilibrium [35].

- We propose a complex traffic modeling framework based on the Active Markov Game, which characterizes dynamic interactions among agents through a joint policy update mechanism, enabling them to effectively learn interactive behaviors in complex environments.
- We propose a game-theoretic interactive training framework that eliminates the "non-responsive opponent" limitation common in conventional simulations, enabling the ego vehicle to indirectly absorb diverse driving strategies (reflecting a variety of driving styles) and to better handle strategic interactions, conflicts, and game-theoretic behaviors among vehicles.
- The proposed framework is implemented and evaluated in the CARLA simulator under unprotected traffic scenarios. Experimental results demonstrate that our method outperforms existing approaches in intersection scenes and exhibits higher stability and safety under complex traffic conditions.

2. Related work

In the early stages of autonomous driving research, motion planning and traditional control methods constituted the core components of the decision-making system. These approaches primarily aimed to generate feasible and safe trajectories from the starting point to the destination, with representative algorithms including RRT, RRT*, and PRM [11, 15, 32, 44, 47]. However, such methods typically rely

on deterministic environmental assumptions, which are difficult to satisfy in real-world scenarios. To address the limitations of traditional methods in complex environments, researchers gradually shifted toward data-driven approaches, particularly the use of deep learning in autonomous driving [12, 14, 29, 30, 43]. Nevertheless, data-driven algorithms remain inherently limited, as they struggle to adequately cover rare and extreme scenarios. To overcome these challenges, recent studies have introduced deep reinforcement learning (DRL) based on simulation platforms [10, 40, 45]. As research progressed, it became increasingly evident that in highly interactive environments such as autonomous driving, single-agent reinforcement learning (RL) is insufficient to capture the intricate game-theoretic interactions and dynamic dependencies among vehicles. To better model the coupling relationships between agents, a number of studies have explored multi-agent reinforcement learning (MARL) [6, 18, 36, 46]. However, despite substantial progress, existing MARL-based methods still suffer from training instability and poor generalization. In particular, under heterogeneous-agent or long-tail extreme scenarios, the simultaneous policy updates of multiple agents introduce environmental non-stationarity, leading to unstable policy dynamics and hindering the convergence of cooperative equilibria within the system.

3. Method

The decision-making algorithm relies on fused sensory inputs, including front-view camera images processed through a Variational Autoencoder (VAE) [17, 39] and a Vision Transformer (ViT), [2, 8] high-definition map information, and radar data that have undergone motion compensation and clustering. A Transformer module implicitly infers the opponent agent's policy and generates actions by sampling from a Gaussian distribution. The environment is modeled under an Active Markov Game framework, where magnetic potential–based reward shaping enables both agents to effectively learn interactive behaviors. Meanwhile, we maintain a policy pool for opponent agents, in which strategies of varying intelligence levels and driving styles are continually added. This provides more realistic interaction experiences and further enhances the decision-making algorithm's generalization capability Fig. 3.

3.1. Active Markov Game Formulation

To explicitly characterize the mutual influence and adaptive behavior among agents, we introduce an **Active Markov Game (AMG)** framework. Unlike the conventional Markov Game (MG), which assumes that both the state transition and reward functions are stationary, AMG allows these functions to dynamically evolve as a function of the agents' policies. Formally, the environment dynamics and reward structure are defined as Eq. 1:

$$s_{t+1} \sim P_U(s_{t+1} \mid s_t, a_t, \pi_t), \quad r_t^i = R_U^i(s_t, a_t, \pi_t), \quad (1)$$

where $a_t = (a_t^1, \ldots, a_t^N)$ and $\pi_t = (\pi_t^1, \ldots, \pi_t^N)$ denote the joint actions and policy profiles of all agents, respectively. This formulation enables the transition kernel P_U and the reward function R_U to explicitly depend on the current strategic configuration π_t, thus capturing the temporal evolution of inter-agent interactions and behavioral adaptation. To illustrate this mechanism, consider a typical merging scenario in autonomous driving, where an ego vehicle interacts with a surrounding opponent vehicle. When the ego vehicle applies a low acceleration, a conventional Markov Game assumes that the next state depends solely on its current speed and control input, ignoring potential reactions from other agents. In contrast, under the AMG formulation, the opponent's strategy π_t^{-i} jointly determines the environmental outcome. If the opponent follows a *cautious* strategy π_t^{safe}, both vehicles decelerate, leading to a safe merging state s_{t+1}^{safe} with positive rewards $r_t^i > 0$. Conversely, if the opponent follows an *aggressive* strategy π_t^{agg}, it may accelerate to block the merge, resulting in a collision state s_{t+1}^{crash} and negative rewards $r_t^i < 0$. This can be formally expressed as Eq. 2:

$$P_U(s_{t+1}^{\text{safe}} \mid s_t, a_t, \pi_t^{\text{safe}}) > P_U(s_{t+1}^{\text{safe}} \mid s_t, a_t, \pi_t^{\text{agg}}), \quad (2)$$

indicating that the same control action a_t may lead to distinct outcomes under different opponent policy distributions.

By incorporating policy-dependent state transitions and rewards, the AMG framework transforms the environment from a passive physical system into a responsive strategic entity, providing a mathematical abstraction for explicitly modeling interactions in decision-making algorithms.

3.2. Sensor Fusion and State Representation

In this section, we describe the process of integrating multi-source perception data and constructing a comprehensive state representation to ensure that the environmental state used for agent decision-making is both accurate and physically meaningful. Our fusion framework incorporates radar data, front-facing camera images, high-precision maps, and vehicle state information (speed, angle, acceleration) to form a unified observation vector for reliable dynamic object detection and tracking in complex traffic scenarios. Radar data, processed through a physics-aware multi-stage pipeline, is first compensated for ego-motion by adjusting for the sensor's translational and rotational velocity components. This compensation removes the apparent motion caused by the vehicle's movement, ensuring that the radar data reflects the actual motion of surrounding objects. After compensation, low-confidence detections (those with low corrected radial velocity or those outside the radar's region

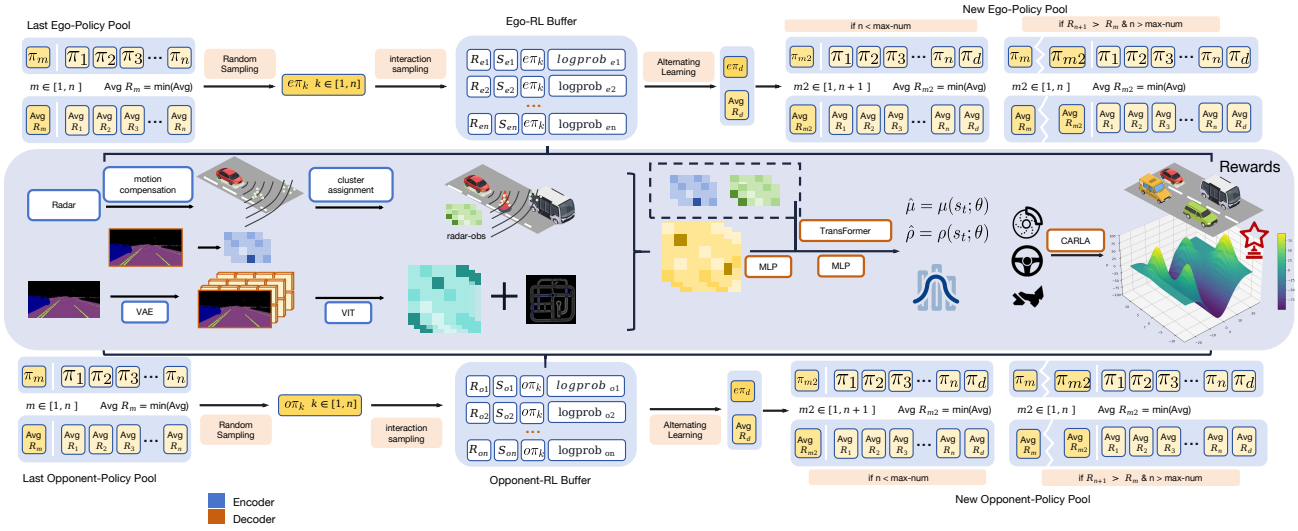


Figure 3. **Overview of the perception and decision-making framework.** The ego vehicle and opponent agents are deployed with heterogeneous decision-making algorithms. For each training batch, the ego agent randomly selects an interaction driving policy with a distinct behavioral style from an opponent policy pool. At every decision step, the algorithm takes as input a sensor-fusion representation that integrates radar signals, front-view camera observations, and high-definition map data, combined with inferred contextual features processed by a Transformer-based module. The output actions are sampled from a Gaussian policy, which are subsequently mapped to steering, throttle, and brake commands. The reward is computed using a magnetic potential field formulation, reflecting the dynamic safety and efficiency of vehicle interactions.

of interest) are discarded. Strong-motion candidates are then subject to a robust motion threshold based on the Median Absolute Deviation (MAD), which suppresses noise under varying signal conditions. Remaining candidates are then optionally clustered in the ground plane to merge multiple reflections from the same object, resulting in stable per-object observations. A multi-frame gating and confirmation process is then applied to maintain lightweight temporal tracks for each candidate, validating them only if consistent across k consecutive frames in azimuth, range, and velocity. In parallel, the vehicle's front-facing camera images are processed using a Variational Autoencoder (VAE) and a Vision Transformer (ViT) to extract features that provide rich, high-dimensional visual information. This processed image data is then fused with the radar information , and combined with vehicle state data (speed, angle, acceleration and historical trajectory) as well as map information. The fusion of radar, vision, and vehicle state data results in a comprehensive and robust representation of the dynamic environment. This sensor fusion and state construction module plays a crucial role in ensuring the reliable detection and tracking of dynamic objects in real-time traffic scenarios. By incorporating physical priors from the radar and maintaining tight coupling between perception and decision-making, our framework enhances the robustness and credibility of the interactive reinforcement learning (RL) process. This approach ultimately provides high-quality input for subsequent game-theoretic decision-making, ensuring that agents make informed, realistic decisions in complex environments.

3.3. Multi-Agent Co-Evolutionary Training Framework

In real-world multi-agent traffic scenarios, decision-making is inherently interactive and non-stationary. The optimal behavior of one vehicle depends not only on environmental conditions but also on the evolving intentions of other agents. To address this challenge, we propose a **Multi-Agent Co-Evolutionary Training Framework**, which integrates game-theoretic principles and reward shaping. This framework enables all agents to co-evolve their strategies through repeated interactions, overcoming the limitations of traditional fixed-opponent training.

The core idea of the framework is to employ a game-theory-driven strategy co-evolution process: all agents evolve their strategies together through repeated interactions, avoiding the fixed opponent model that has limited adaptability. Each agent maintains a **policy pool** $\mathcal{P}^i = \{\pi_1^i, \pi_2^i, \ldots, \pi_K^i\}$, where each element represents a strategy learned in a previous training phase. At the beginning of each training episode, the opponent agent samples a policy from its pool either randomly or according to a distribution over the opponent's policies. This provides the ego agent with a diverse and adaptive interaction environment, pre-

venting overfitting to a single opponent model. The mixed strategy is represented as:

$$\bar{\pi}^{-i} = \sum_k w_k \pi_k^{-i}, \quad \text{s.t.} \sum_k w_k = 1. \tag{3}$$

For each agent, the training objective is to learn a **best-response policy** π^i_{BR} to the sampled opponent mixture $\bar{\pi}^{-i}$, maximizing the expected return under the stochastic multi-agent transition dynamics:

$$\max_{\theta^i} \mathbb{E}_{s_t, a_t \sim (\pi^i_{\text{BR}}, \bar{\pi}^{-i})} \left[\sum_{t=0}^{T} \gamma^t r^i(s_t, a_t) \right]. \tag{4}$$

In continuous control settings, Proximal Policy Optimization (PPO) is used to stabilize training. Once the ego agent's policy converges, the new policy is added to the policy pool, thereby enhancing the diversity of strategies in the agent population. Over time, this iterative process gradually drives the system towards a game-theoretic equilibrium.

To enhance the training process, a **magnetic potential field-based reward mechanism** is introduced. In this mechanism, the ego-road interaction is treated as an attractive potential, encouraging the ego vehicle to adhere to the navigation centerline, while ego-agent interactions are modeled as repulsive potentials to avoid collisions. The reward function incorporates the safety distance and relative velocity between agents, providing dense feedback signals that address sparse reward issues and promote stable training. The reward shaping function is defined as:

$$\begin{aligned} r_{\text{shape}} = {} & w_{\text{int}} \left(\Phi_{\text{int}}(t_n) - \Phi_{\text{int}}(t_{n-1}) \right) \\ & + w_{\text{lane}} \left(\Phi_{\text{lane}}(t_n) - \Phi_{\text{lane}}(t_{n-1}) \right) \end{aligned} \tag{5}$$

where Φ_{int} and Φ_{lane} are the interaction and lane potential terms, respectively, and are computed as follows:

$$\Phi_{\text{int}} = w(d) \left(\alpha U_d(d) + \beta U_{\text{TTC}} \right), \tag{6}$$

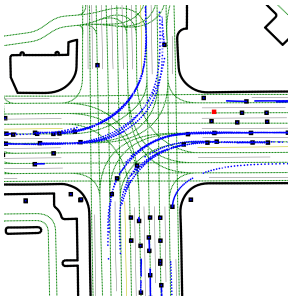
$$U_d(d) = \frac{1}{d^{k_d}}, \quad \text{TTC} = \frac{d}{\max(c, \varepsilon)}, \tag{7}$$

$$\Phi_{\text{lane}} = \frac{1}{2} \left(\left(\frac{d}{d_{\max}} \right)^2 + \left(\frac{|\Delta\psi|}{\psi_{\max}} \right)^2 \right), \tag{8}$$

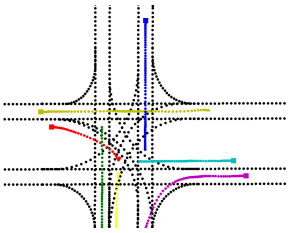
where d is the distance to the target, $\Delta\psi$ is the heading deviation, and TTC is the time-to-collision. These terms allow the agent to learn to maintain a safe distance from other vehicles and the road centerline, improving safety and efficiency.

4. Experiments

In the experiments, we first conducted a visualization analysis on publicly available datasets. Fig. 4a The results show that these datasets feature well-structured traffic facilities and compliant driving behaviors, with almost no occurrences of aggressive interactions such as lane competition or abrupt merging. Consequently, the data distribution provides insufficient information for learning effective interaction strategies.Next, we visualized the decision-making process of the rule-based agents built into CARLA. Fig. 4b It can be clearly observed that these agents simply follow predefined routes and stop when obstacles are detected, without exhibiting any interactive behaviors. Such rule-driven trajectories fail to reflect the complexity and uncertainty of real-world multi-agent traffic interactions.To further demonstrate the effectiveness of our proposed algorithm, we conducted a series of experiments, including comparative studies, ablation experiments, hyperparameter sensitivity analyses, and opponent agent capability evaluations, to comprehensively assess the algorithm's performance and adaptability in various interactive scenarios.



(a) Visualization of Trajectories in the Waymo Dataset



(b) Visualization in the CARLA Dataset

Figure 4. Trajectory visualizations in two datasets

4.1. Experimental Setup

All experiments are conducted using high-performance GPUs, with algorithm training performed on **NVIDIA A100** GPUs. The experimental framework is built on the high-fidelity simulation platform **CARLA**.

During the training phase, we utilize the Town10HD and Town02 maps provided by **CARLA**, where we configure multiple unprotected intersections, T-junctions, left and right turns Fig. 5a, and routes with significant spatiotemporal conflicts, forming long-tail scenarios that require vehicles to perform interactive decision-making to navigate safely. These complex settings simulate conditions without traffic signals or lane markings, representing situations where the lack of traffic control facilities can lead to potential collisions. The overall configuration of the experimental environments is illustrated in Figs. 5(a) and (b).

All test scenarios are two-vehicle settings and are configured with traffic-signal-free intersections and conflict-rich routes. The test set also includes roundabout and merging scenarios that are unseen during training, Fig. 5b to assess the generalization capability of the trained policy in novel interactive environments.

During training, three random seeds (0, 45, and 2025)



(a) Visualization of interactive behaviors in training scenarios.



(b) Visualization of interactive behaviors in testing scenarios.

Figure 5. Visualization of scenarios.

are used to ensure reproducibility and robustness. For each scenario, each strategy is trained for approximately 100,000 interaction steps. In the testing phase, 100 repeated evaluations are conducted for each random seed, and the averaged results are reported.

For all non-hyperparameter experiments, the default parameter settings are as follows: the interaction distance R_{int} is set to 17, the sensitivity of vehicles to distance k_d is 2.0, the obstacle rejection coefficient α_{rep} is 0.5, the time-to-collision potential intensity β_{ttc} is 0.5, the interaction weight w_{int} is 0.6. The prediction horizon is set to one step.

4.2. Analysis of Comparative Performance

Tab. 1 presents the quantitative comparison of various reinforcement learning and multi-agent baselines under conflict-rich scenarios designed without traffic control facilities. These settings require both agents to engage in interactive decision-making to safely navigate through intersections or merging zones.

In the single-agent setting, the ego vehicle interacts with rule-based opponent agents or multi-style decision-making agents. When trained against rule-based opponents, traditional methods such as DQN, PPO, and DDPG achieve moderate performance in terms of return and collision avoidance but exhibit limited adaptability to diverse behaviors. These agents often overfit to the deterministic motion patterns of the rule-based opponents, resulting in suboptimal generalization to unseen situations. When the opponents are replaced with multi-style decision-making agents, the performance of most single-agent algorithms

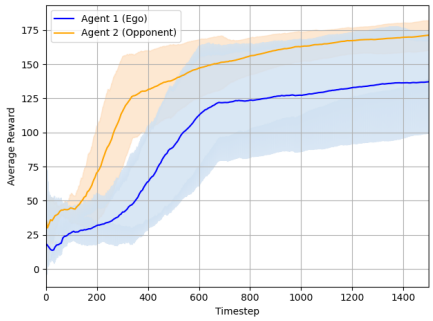


Figure 6. Average Reward of Ego and Opponent Agents During Training in a Single Strategy

decreases noticeably, particularly in safety-related metrics such as Collision Rate and Risk Time. This highlights their vulnerability to the increased behavioral variability of opponents. In contrast, our proposed interaction-aware framework maintains stable performance, demonstrating stronger robustness and adaptability under dynamic and uncertain interactions.

In the multi-agent setting, all agents learn their policies simultaneously, allowing cooperative or competitive behaviors to emerge naturally. Baselines such as IPPO outperform single-agent approaches in terms of safety and efficiency, confirming the advantage of centralized training with decentralized execution in multi-agent frameworks. However, although these multi-agent reinforcement learning baselines can effectively model interactive decision-making, the opponent agents still lack sufficient behavioral diversity and typically exhibit only a single, fixed interaction pattern, often resorting to conservative behaviors such as stopping and yielding in conflict scenarios. As a result, the ego agent can often achieve a high success rate by adopting relatively simple conservative strategies, such as slow driving, without learning more sophisticated game-theoretic interaction behaviors. This training bias limits the richness, adaptability, and generalization of the learned policy in complex traffic conflicts.

When our proposed game-theoretic interaction training algorithm is applied—where each opponent maintains a diverse policy pool and randomly samples one for decision-making during training—the ego agent is exposed to a wide range of behavioral strategies. This diversity significantly enhances its ability to anticipate and adapt to different driving intents, resulting in better generalization and robustness. Our method achieves the lowest Collision Rate (0.02) and Path Deviation (0.8652), while attaining the highest Return (138.9278). The improvement in Safety Margin and Control Smoothness further indicates that the learned policy can handle interactive conflicts safely and smoothly without sacrificing efficiency Fig. 6.

Overall, these results demonstrate that training with diverse opponent behaviors leads to a richer and more robust

Table 1. Comprehensive comparison of algorithms under different opponent settings in both single-agent and multi-agent frameworks.

Framework	Method	Collision	Risk Time	Return	Path Deviation	Safety Margin	OOD Success Rate	Control Smoothness
Single-agent (Rule-based)	DQN [25, 34]	0.32	2.35	93.4672	1.2343	0.52	0.61	0.28
	PPO [27, 41]	0.08	1.22	132.9834	0.9864	**3.65**	0.92	0.44
	DDPG [21, 22]	0.13	2.13	120.4325	1.1203	2.76	0.85	0.35
Single-agent (Diverse Opponent)	DQN	0.68	2.12	62.6754	1.2412	0.22	0.32	0.31
	PPO	0.60	2.41	68.3281	0.9324	0.41	0.40	0.32
	DDPG	0.54	2.43	70.3123	1.0414	0.26	0.46	0.29
Multi-agent (Rule-based)	IPPO [7]	0.03	1.92	132.5123	0.87623	2.13	0.97	0.42
	Ours (Potential)	**0**	**1.02**	**138.9278**	**0.8652**	3.17	**1**	**0.45**
Multi-agent (Diverse Opponent)	IPPO	0.29	2.88	126.3025	0.9423	1.15	0.71	0.39
	Ours (Potential)	**0.02**	**1.93**	**133.7421**	**0.8821**	**2.12**	**0.98**	**0.43**

interaction policy. By maintaining a strategy pool for opponent agents and sampling from it probabilistically during training, our method effectively simulates the variability of real-world drivers. This enables the ego vehicle to develop more generalized, adaptive, and collision-resistant decision-making capabilities in complex, unregulated traffic environments Fig. 7.

4.3. Analysis of Hyperparameter Sensitivity

Through sensitivity analysis of the four hyperparameters (R, k_d, w_{int}, and β) in Table 2, we observe several consistent performance trends. As R increases, the best performance in terms of both return and OOD success rate is achieved when R is in the range of 15–17 m, whereas a larger radius of around 20 m improves the safety margin but reduces the return. Increasing k_d generally improves both return and OOD success rate, with the strongest results obtained when k_d is between 1.5 and 2.0; however, overly large decay factors (e.g., $k_d \approx 2.5$) slightly degrade the return. The interaction weight w_{int} affects both strategy diversity and safety: overall performance is highest when w_{int} is between 0.4 and 0.6, while larger interaction weights (around 0.7) further enhance safety but lead to lower return and weaker generalization. It is worth noting that, as β increases, both return and OOD success rate first improve and reach their best levels within a moderate range (approximately 0.5–0.6). However, larger values (e.g., around 0.9) make the agent overly aggressive in collision avoidance, causing it to adopt extreme maneuvers, such as leaving the lane, in order to evade interacting vehicles, which in turn degrades both the overall return and safety. Overall, moderate hyperparameter choices tend to provide a better balance across metrics, and careful tuning substantially improves robustness and generalization.

4.4. Analysis of Strategy Diversity

We evaluate three types of algorithms based on action diversity: Rule-based Algorithm, Single-Agent Training, and Interactive Training. Tab. 3 We use Strategy Diversity and Action Space Coverage as evaluation metrics. To quantify strategy diversity, we measure the degree of variation among different strategies. Specifically, the action sequence of each episode is regarded as a representation of the underlying strategy. We then compute the cosine dissimilarity between every pair of action sequences, defined as $1 - \frac{a_i^\top a_j}{\|a_i\|\|a_j\|}$, and average it over all pairs to obtain the final strategy diversity score. A higher score indicates larger behavioral differences among strategies and therefore greater strategy diversity. The Rule-based Algorithm achieved a Strategy Diversity of 0.1023, indicating that its strategies are relatively simple and lack diversity. Its Action Space Coverage was 0.1547, suggesting that it covers only a limited portion of the action space. Action Space Coverage measures the ratio of the number of distinct actions selected by the agent during training to the total size of the action space. Single-Agent Training showed improvement with a Strategy Diversity of 0.3278 and Action Space Coverage of 0.4125, reflecting greater adaptability and broader action space coverage compared to the Rule-based Algorithm. The Interactive Training performed the best, with a Strategy Diversity of 0.5841 and Action Space Coverage of 0.6394, demonstrating significant advantages in generating diverse

Table 2. Hyperparameter sensitivity (Rule-based). Each hyperparameter is varied independently and default parameters are listed in the main table.(The safety factor is defined as the sum of the safety margin and the signed distance to the lane boundary, minus the risk time, where the distance is positive when the vehicle remains within the lane and negative once it crosses the lane boundary.)

Hyperparameter (Value)	Avg. Return ↑	OOD Succ. ↑	Safety Factor ↑
$R = 10\,\text{m}$	132.6593	0.91	3.23
$R = 15\,\text{m}$	**139.5102**	0.99	3.74
$R = 20\,\text{m}$	134.1992	**1**	**4.93**
$k_d = 1.0$	133.0847	0.90	3.09
$k_d = 1.5$	**138.2976**	**0.98**	**3.83**
$k_d = 2.5$	134.1165	0.96	3.55
$w_{int} = 0.2$	130.9524	0.89	3.19
$w_{int} = 0.4$	**135.3574**	0.94	3.78
$w_{int} = 0.7$	128.1268	**1**	**3.84**
$\beta = 0.4$	118.3134	0.92	3.85
$\beta = 0.6$	**128.0711**	**0.98**	**3.97**
$\beta = 0.9$	106.6458	0.79	2.96

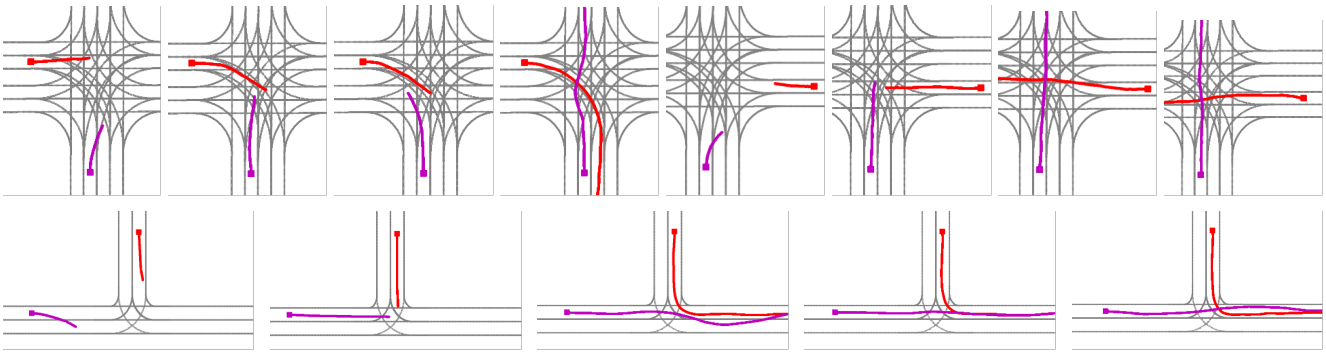


Figure 7. Visualization of the training process. The agent gradually learns to handle different interactive scenarios: the upper-left panels show oncoming interactions, the upper-right panels illustrate lateral interactions, and the bottom sequence depicts same-direction interactions. The last three bottom panels present different strategies under the same situation, demonstrating the policy's adaptability and generalization through continuous interactive learning.

Table 3. Comparison of strategy diversity and action coverage across different algorithms.

Algorithm	Strategy Diversity ↑	Action Space Coverage ↑
Rule-based Algorithm	0.1023	0.1547
Single-Agent Training	0.3278	0.4125
Interactive Training	**0.5841**	**0.6394**

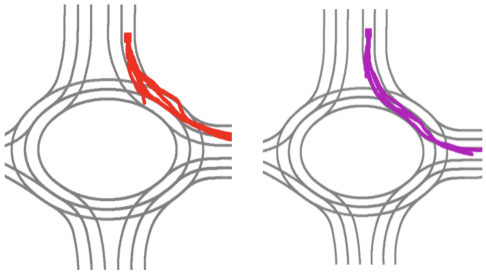


Figure 8. Visualization of Roundabout Navigation Strategies

strategies and exploring a wider range of actions, allowing it to better adapt to complex environmental changes Fig. 8.

4.5. Analysis of Ablation Study

In this ablation study Tab. 4, we systematically remove different modules to assess their impact on model performance. The results show that the full model performs best in both OOD success rate and crash rate, with an OOD success rate of 0.98 and a crash rate of 0.02. This indicates that the full model successfully handles variations in the environment with minimal collisions, demonstrating high stability and safety. After removing the strategy inference module, the OOD success rate drops to 0.84, while the crash rate increases to 0.15, suggesting that this module plays a crucial role in enabling the model to adapt to different distributions and ensuring safety. When the strategy pool module is removed, the OOD success rate further decreases to 0.81, and the crash rate increases to 0.17, highlighting the importance of this module in improving the model's adaptability

Table 4. Ablation study of the proposed method under the multi-agent setting.

Configuration	OOD Success Rate ↑	Collision ↓
Full Model (Ours)	**0.98**	**0.02**
w/o Strategy Inference Module	0.84	0.15
w/o Strategy Pool Module	0.81	0.17
w/o Map Module	0.62	0.23
w/o Perfect Perception Module	0.54	0.46

and decision stability. Removing the map module results in a significant drop in OOD success rate to 0.62 and a notable increase in the crash rate to 0.23, demonstrating the critical role of the map module in environmental perception and collision avoidance. Finally, after removing the perfect Perception module, the OOD success rate decreases to 0.54, and the crash rate increases to 0.46, indicating the importance of this module in the model's comprehensive understanding of the environment and safe decision-making. In summary, all modules contribute to the model's performance, and removing any of them leads to noticeable drops in OOD success rate and safety.

5. Conclusion

The proposed Active Markov Game-based multi-agent framework significantly enhances autonomous driving decision-making in complex, unsignalized scenarios. By modeling policy-dependent dynamics and co-evolving agent strategies, our approach effectively captures the uncertainty and interactions in real-world traffic. Experimental results demonstrate that our system learns adaptive strategies that traditional rule-based or single-agent methods cannot achieve, providing a robust, safe, and adaptive decision-making framework for autonomous driving in traffic environments without signal control.

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